Budget Beliefs:

Testing Bias in Perceived Spending Habits

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Introduction

Introduction

What are we investigating?

- How perceptions of student spending are influenced by **gender** and **budget level**.
  - Do participants allocate more to **essentials** for lower-budget students, and more to **non-essentials** for higher-budget students?
  - Are allocations different on **perceived male** student or **perceived female** student?

Introduction

Why does this matter?

- **Perceptions of spending** can influence how we judge someone's **financial responsibility** and **spending priorities**
    - Real world example: **Pink Tax**, "women products" (like shampoo, razors, etc.) cost more than similar products for men due to biased assumptions about spending
- Understanding these biases helps us identify **financial inequalities** and challenge them before they shape broader economic decisions

Experimental Design

Experimental Design

Pre-Test:

Average Monthly Spending?



X 20

Monthly Spending:

Min :	**$1,050**
Average:	**$1,700**
Max:	**$2,200**

Experimental Design

Budget Level / F/M name	Low $1,050	Medium $1,700	High $2,200
F			
M		P - Value	P - Value

2 x 3 Factorial Design

Reference group (held as baseline in comparison)

+ Block on Gender of Participant

Experimental Design

Total Participants

60

Surveys

6

Participants per version

10 (5M, 5F)

Survey Structure and Sampling

	Budget Level $1,050	Budget Level $1,700	Budget Level $2,200
Sara	5M | 5F	5M | 5F	5M | 5F
Mike	5M | 5F	5M | 5F	5M | 5F

Experimental Design

Full survey sample can be seen in appendix at the end of the video

Name:	Madison
Gender:	Female

Create a monthly budget for your classmate Christopher!

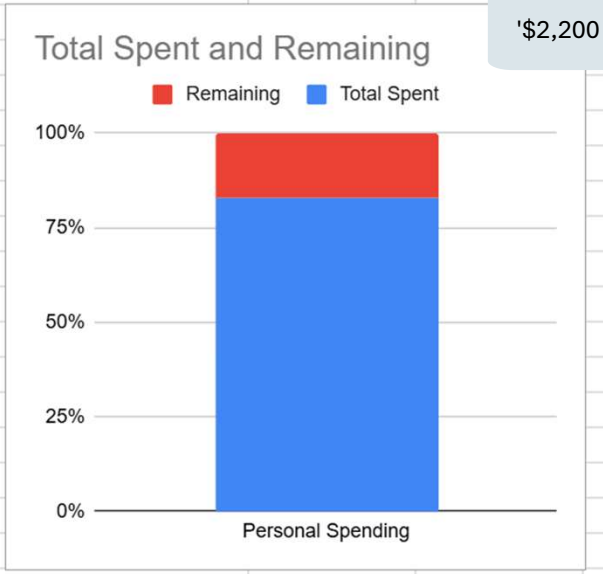
Listed below are various categories of typical spending items for undergrad students, categorized as essential or non-essential.

Fill in the dollar amounts as you see fit of how much you think Christopher should allocate to each category (some can be kept as 0)

Note that he wants to spend under **$2,200** a month!

Item	**Category**	***insert here ($)***
Takeout/Restaurant	Non-Essential	$100.00
Groceries	Essential	$300.00
Online subscriptions	Non-Essential	$50.00
Transportation (gas/public transit)	Essential	$40.00
Phone plan	Essential	$45.00
Uber/Lyft	Non-Essential	$20.00
Gym membership	Non-Essential	$50.00
School supplies	Essential	$20.00
Rent/utilities	Essential	$1,200.00
Shopping (apparel, skin care)	Non-Essential	
Social entertainment (hobbies/events/fun stuff)	Non-Essential	
Health (out of pocket expenses, mental health)	Essential	

	Total Spent	**Remaining**
Personal Spending	$1,825.00	$375.00



Name filled out -> Participant

Gender filled out -> Blocking Variable

'Christopher' -> Factor 2

'$2,200 Budget' -> Factor 1

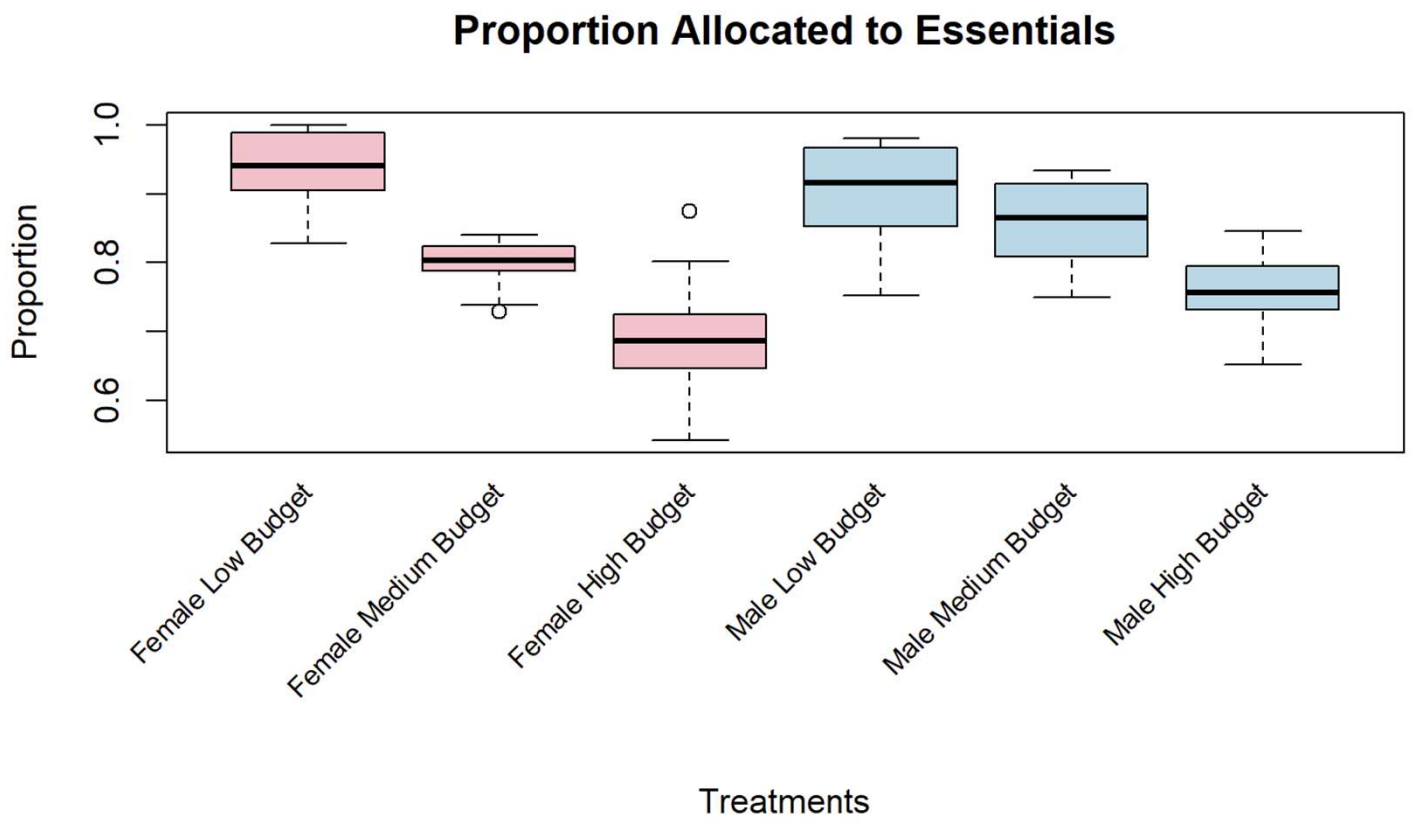
Experimental Design

	Blocking Variable	Factor 1	Factor 2	Y-variable
ID	**Gender of Participant**	**Budget Level**	**Gender of Name**	**% Spent on Essentials**
1	Male	1050	Male	0.9143
2	Female	1700	Male	0.8035
…	…	…	…	…
59	Male	2200	Female	0.7572
60	Female	1700	Female	0.8359

Sample Data

Exploratory Analysis

Analysis:


Proportion Allocated to Essentials
Proportion
1.0
0.8
0.6
Female Low Budget
Female Medium Budget
Female High Budget
Male Low Budget
Male Medium Budget
Male High Budget
Treatments

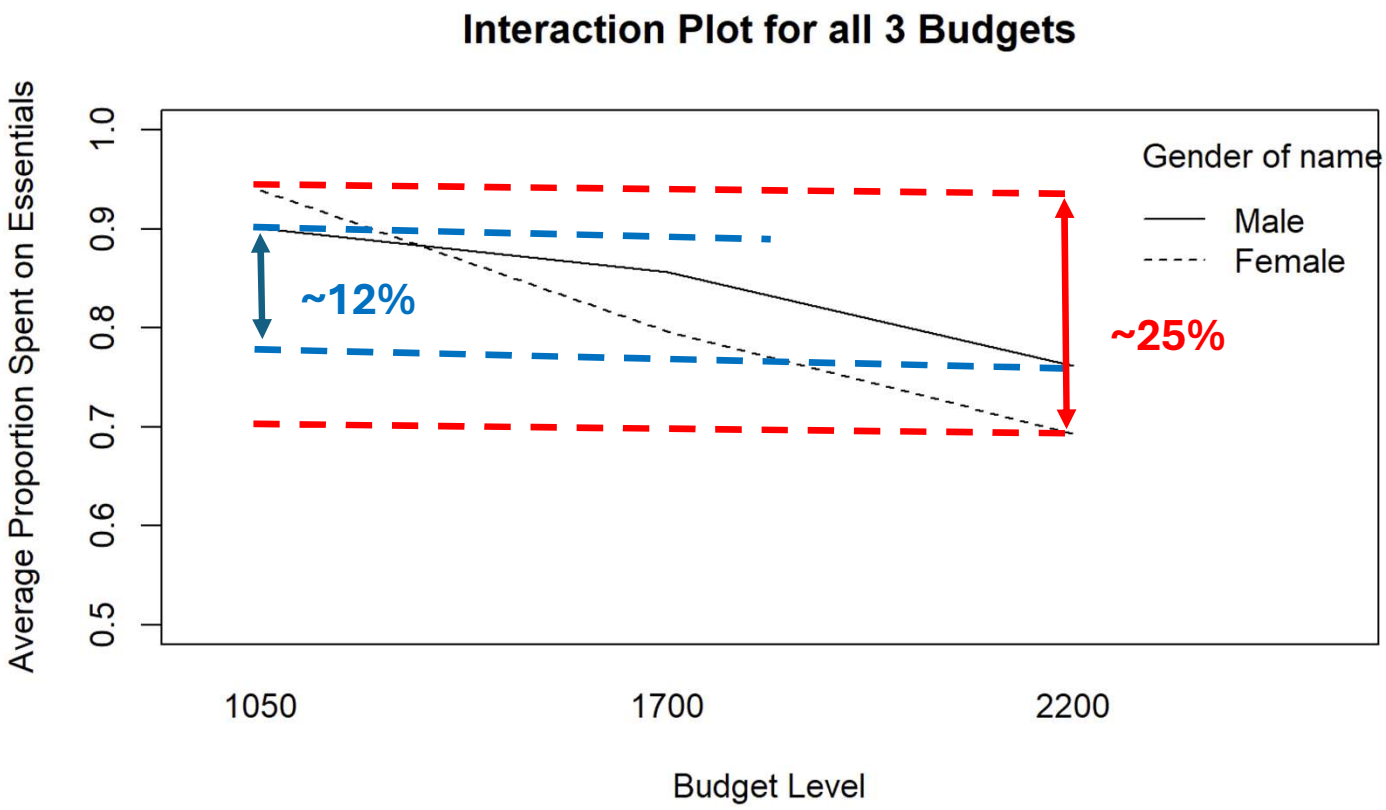
Analysis:

```
Analysis of Variance Table

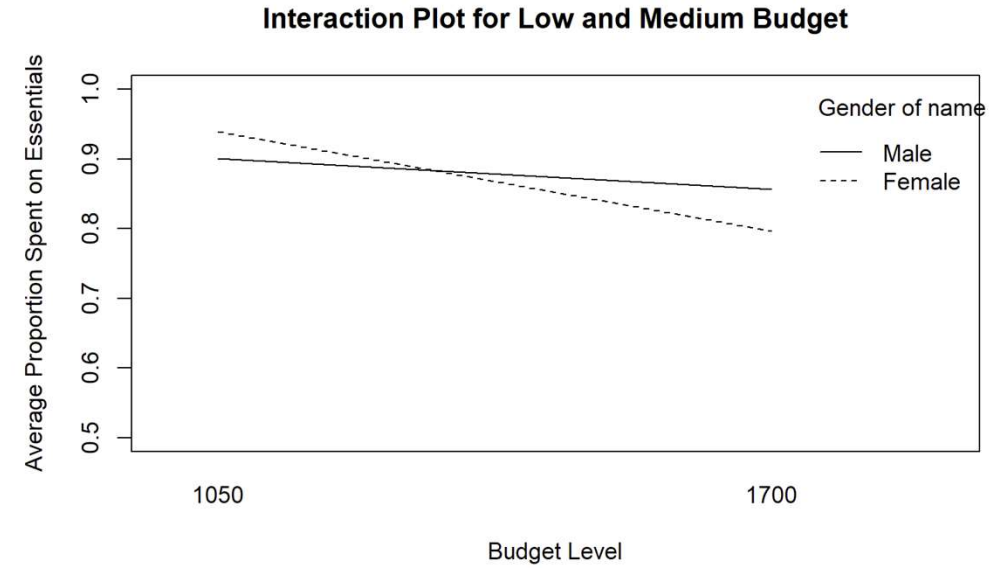
Response: df$`% spent on essential`
                Df  Sum Sq  Mean Sq F value    Pr(>F)
factor1          2 0.34366 0.171830 35.6758 2.037e-10 ***
factor2          1 0.01082 0.010817  2.2459   0.14013
Block            1 0.00180 0.001799  0.3736   0.54377
factor1:factor2  2 0.03276 0.016378  3.4005   0.04108 *
Residuals       51 0.24564 0.004816
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

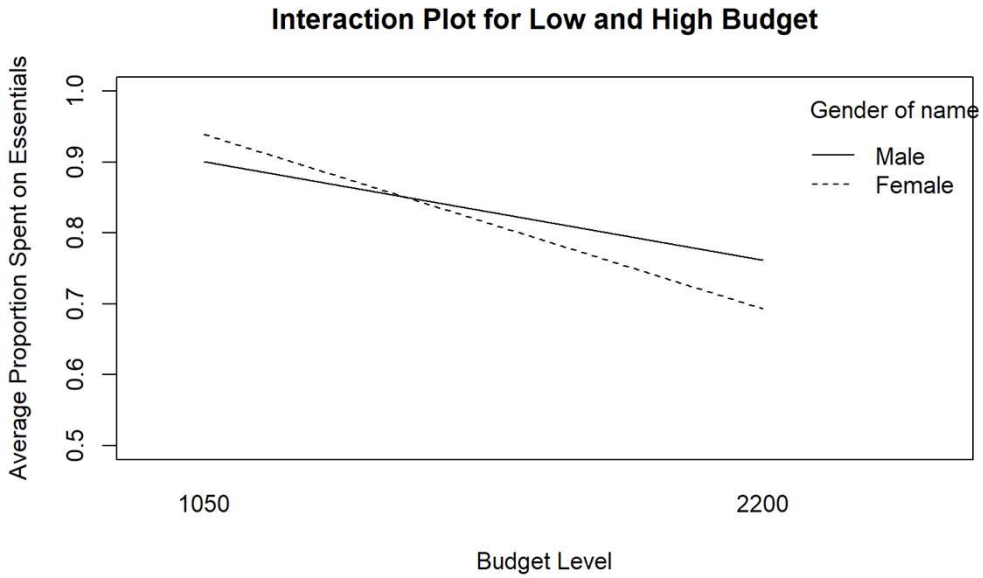
*Factor 1: Budget Levels
*Factor 2: Gender Perception
*Factor 1 : Factor 2: Interaction between factor 1 and 2
*$\alpha = 0.05$

Analysis


Interaction Plot for all 3 Budgets
Average Proportion Spent on Essentials
0.5
0.6
0.7
0.8
0.9
1.0
1050
1700
2200
Budget Level
Gender of name
Male
Female
~12%
~25%

Analysis


Interaction Plot for Low and Medium Budget
Average Proportion Spent on Essentials
0.5
0.6
0.7
0.8
0.9
1.0
1050
1700
Budget Level
Gender of name
Male
Female


Interaction Plot for Low and High Budget
Average Proportion Spent on Essentials
0.5
0.6
0.7
0.8
0.9
1.0
1050
2200
Budget Level
Gender of name
Male
Female

Conclusion

Limitations to our Experiment

- Insignificant blocking factor : `Gender of name`
- Relatively small sample size compared to our population
- Classification of Essential and Non-essential was made on our interpretation
  - Could leave participants to freely classify items as Essential/Non-Essential

Extensions to our Experiment

- Block: International vs domestic students
- Adding contextual scenarios
  - Adding a vignette on student's situation and measure how context changes allocations

Thank you

Appendix

Hello fellow Warrior!

We are a group of 3 FARM/STAT students trying to uncover how budget awareness affects student spending! As students, we all know what it's like to balance limited money with endless expenses—food, supplies, outings, and more. In this experiment, we're exploring whether being more aware of your budget actually changes how you spend. Do you cut back? Spend smarter? Or does it not make much of a difference? Let's find out together!

Thank you for your participation in our final project for STAT 332: Sampling and Experimental Design.

From,
Madison, Raymond and Darren.

Name:	Madison
Gender:	Female

Create a monthly budget for your classmate Christopher!

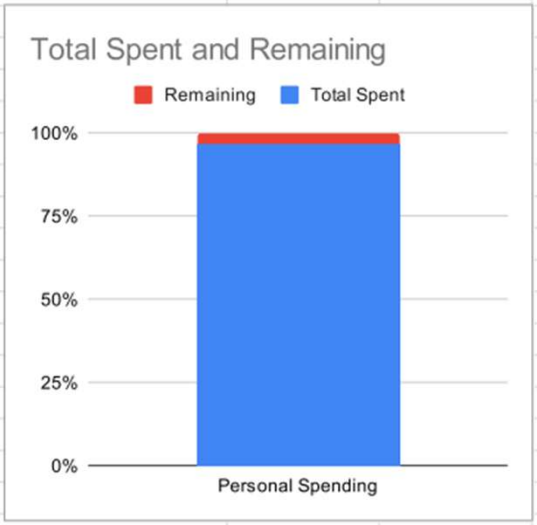
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Fill in the dollar amounts as you see fit of how much you think Christopher should allocate to each category (some can be kept as 0)

Note that he wants to spend under **$2,200** a month!

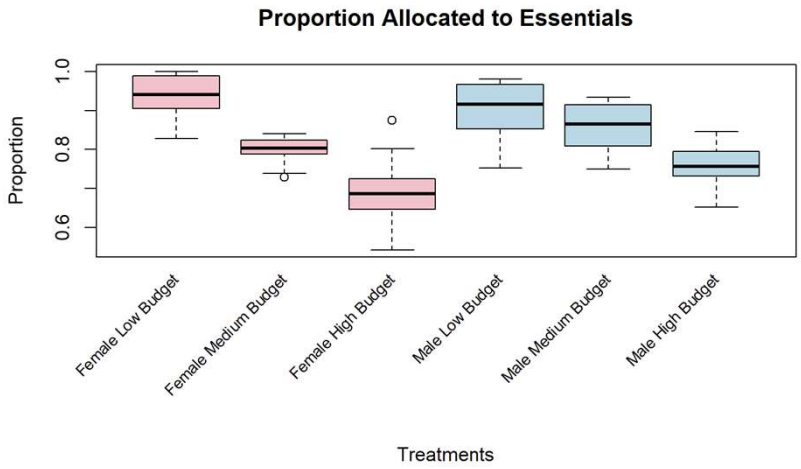
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Phone plan	Essential	$45.00
Uber/Lyft	Non-Essential	$20.00
Gym membership	Non-Essential	$50.00
School supplies	Essential	$20.00
Rent/utilities	Essential	$1,200.00
Shopping (apparel, skin care)	Non-Essential	$150.00
Social entertainment (hobbies/events/fun stuff)	Non-Essential	$100.00
Health (out of pocket expenses, mental health)	Essential	$50.00

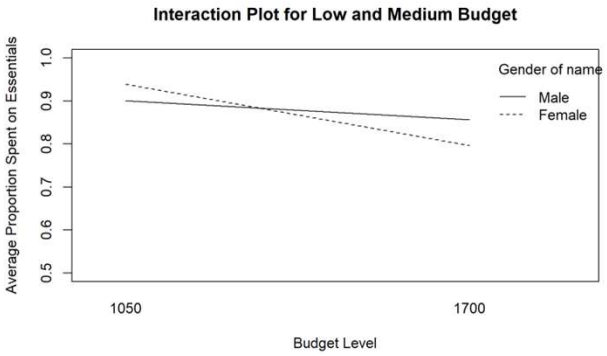
	Total Spent	**Remaining**
Personal Spending	$2,125.00	$75.00

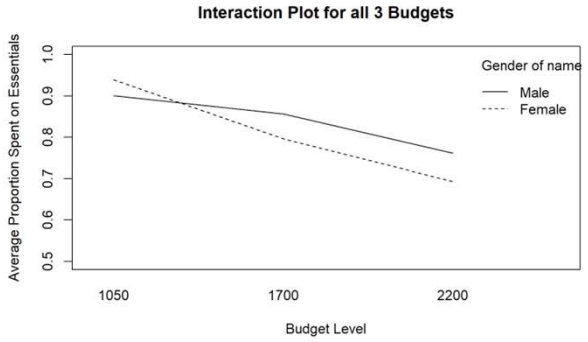


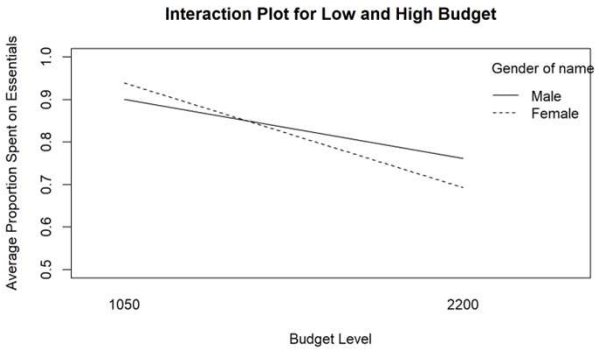
```
Analysis of Variance Table

Response: df$`% spent on essential`
          Df  Sum Sq  Mean Sq F value    Pr(>F)
Block      1 0.00089 0.000886  0.1687    0.6829
factor1    2 0.34457 0.172286 32.7995 5.372e-10 ***
factor2    1 0.01082 0.010817  2.0594    0.1571
Residuals 53 0.27839 0.005253
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```




Interaction Plot for Low and Medium Budget
Average Proportion Spent on Essentials
0.5
0.6
0.7
0.8
0.9
1.0
1050
1700
Budget Level
Gender of name
Male
Female


Interaction Plot for all 3 Budgets
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1050
1700
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Budget Level
Gender of name
Male
Female


Interaction Plot for Low and High Budget
Average Proportion Spent on Essentials
0.5
0.6
0.7
0.8
0.9
1.0
1050
2200
Budget Level
Gender of name
Male
Female